Week 4 Part B — Smoothing, windows and panel shape

Dataset: sessions_hourly.csv (synthetic CDB601220 course data). Daily session totals are hourly flow counts summed by day.

Cleaning and audit

Parsed mixed timestamp formats; removed duplicate timestamps (one logger copy retained); removed sessions = -1 sentinels; summed hourly sessions to daily totals. The extract ends 11 September 2025, so the partial September 2025 month is excluded. Analysis period: 1 April 2024–31 August 2025 (518 calendar days). There are 13 dates with no valid hourly observations. Those dates remain missing in the window report and cleaned CSV; they are time-interpolated only for methods that require a complete series.

1. Window report

7-day: raw SD 6,794.1353; smoothed SD 4,593.4357; 32.4% variation removed; 3 periods lost at each end. 28-day: smoothed SD 4,367.1416; 35.7% removed; 14 periods lost at each end. 91-day: smoothed SD 4,091.3692; 39.8% removed; 45 periods lost at each end. A 7-day window cancels the weekday/weekend cycle. Use a centred 7-day mean in Part D as the weekly seasonal reference, retaining the raw daily line to preserve events. A 91-day window instead suppresses quarter-scale movement and is a different hypothesis, not just a smoother chart.

2. Three smoothers against Black Friday

On 29 November 2024, raw sessions = 113,710; neighbourhood median (22 Nov–6 Dec) = 24,625. Centred 14-day mean = 32,871; centred 14-day median = 24,698; causal EWMA (span=14, adjust=False) = 37,517.75. For a live dashboard, use a causal EWMA alongside the raw series because it does not use future observations. For a quarterly report focused on typical levels, use the centred median and annotate/show the raw Black Friday spike separately.

3. Seasonal decomposition

Additive decomposition with period=7 gives weekly seasonal peak-to-trough amplitude 7,324.20 sessions/day. Largest positive residual: 29 November 2024, +75,029.08 sessions/day, coinciding with Black Friday. The 13 entirely missing dates were time-interpolated only for decomposition. The requested period=365 run fails because seasonal_decompose requires 730 observations (two annual cycles), but only 518 days are available. With enough annual coverage, a repeatable annual pattern would move from resid into seasonal; this extract is too short to estimate that reliably.

4. Trailing or centred

For a 31-day mean, centred minimum = 15 May 2024; trailing minimum = 30 May 2024; lag = 15 days, equal to the half-window. A live dashboard must show a trailing window because centred smoothing uses future values. Suggested caption: “Trailing 31-day mean of daily sessions; uses today and the prior 30 days; turning points may appear about 15 days late relative to a centred estimate.” Keep the raw series visible.

5. The window is a claim

Sensitivity setup: 12 start months, April 2024 through March 2025; for each, fit an OLS line to the first 7, 14, 28, 56 or 91 daily values from the month's first day. Result: 34 rising slopes and 26 falling slopes out of 60. The direction depends on endpoints and window length. Tufte’s Lie Factor assumes a defined effect in the data and a corresponding effect encoded by the graphic. A cherry-picked window changes the reference effect itself, so the denominator is not fixed and a meaningful Lie Factor cannot be computed. Honest substitute: disclose the range/window, show the full raw context, and report window sensitivity (34 rising, 26 falling).

Reproducibility

Run: python -m pip install pandas numpy statsmodels matplotlib; then python partB_analysis.py. CSV results are in results/. All data are synthetic course data, not findings about a real business.